

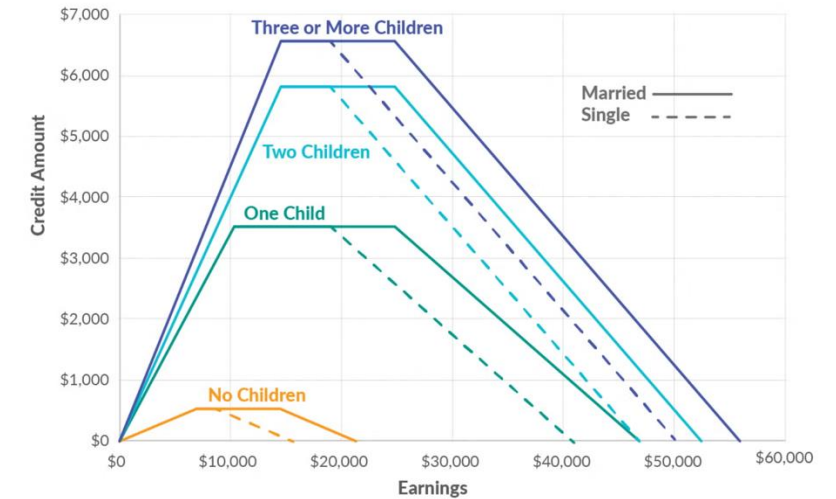
Econ 139: Labor Economics

Labor Supply (Empirics)

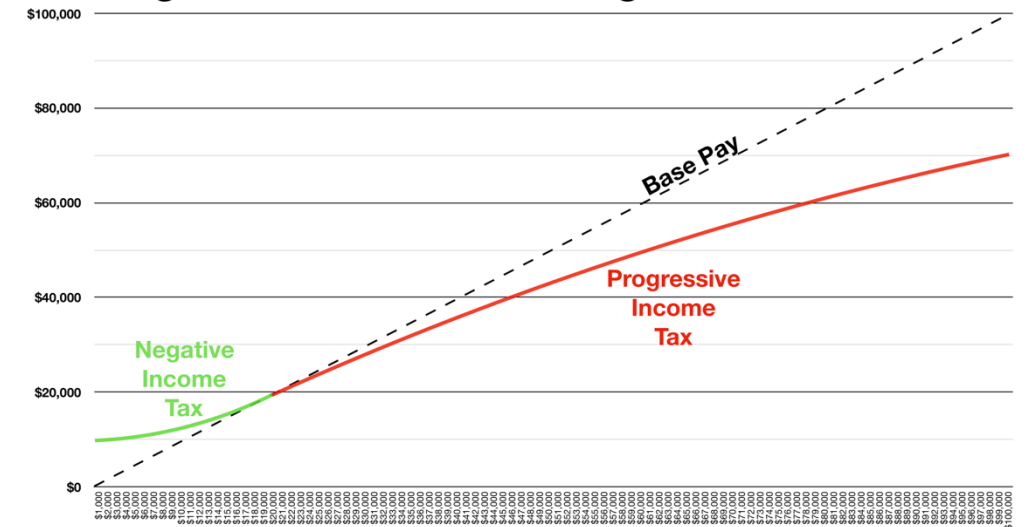
David Arnold

The Phase-In and Phaseout of the EITC

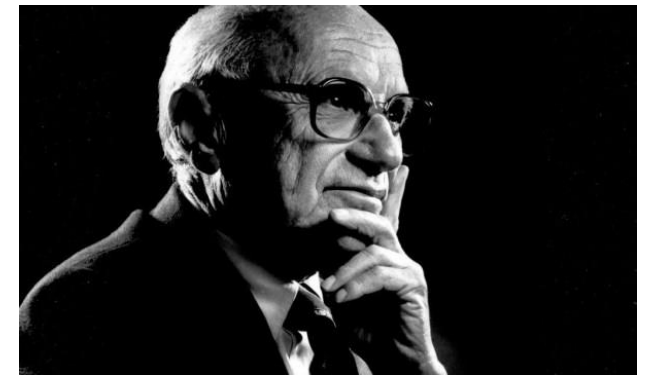
Credit Amount by Marital Status and Number of Children



Negative Income Tax & Progressive Income Tax



Negative Income Tax Experiments



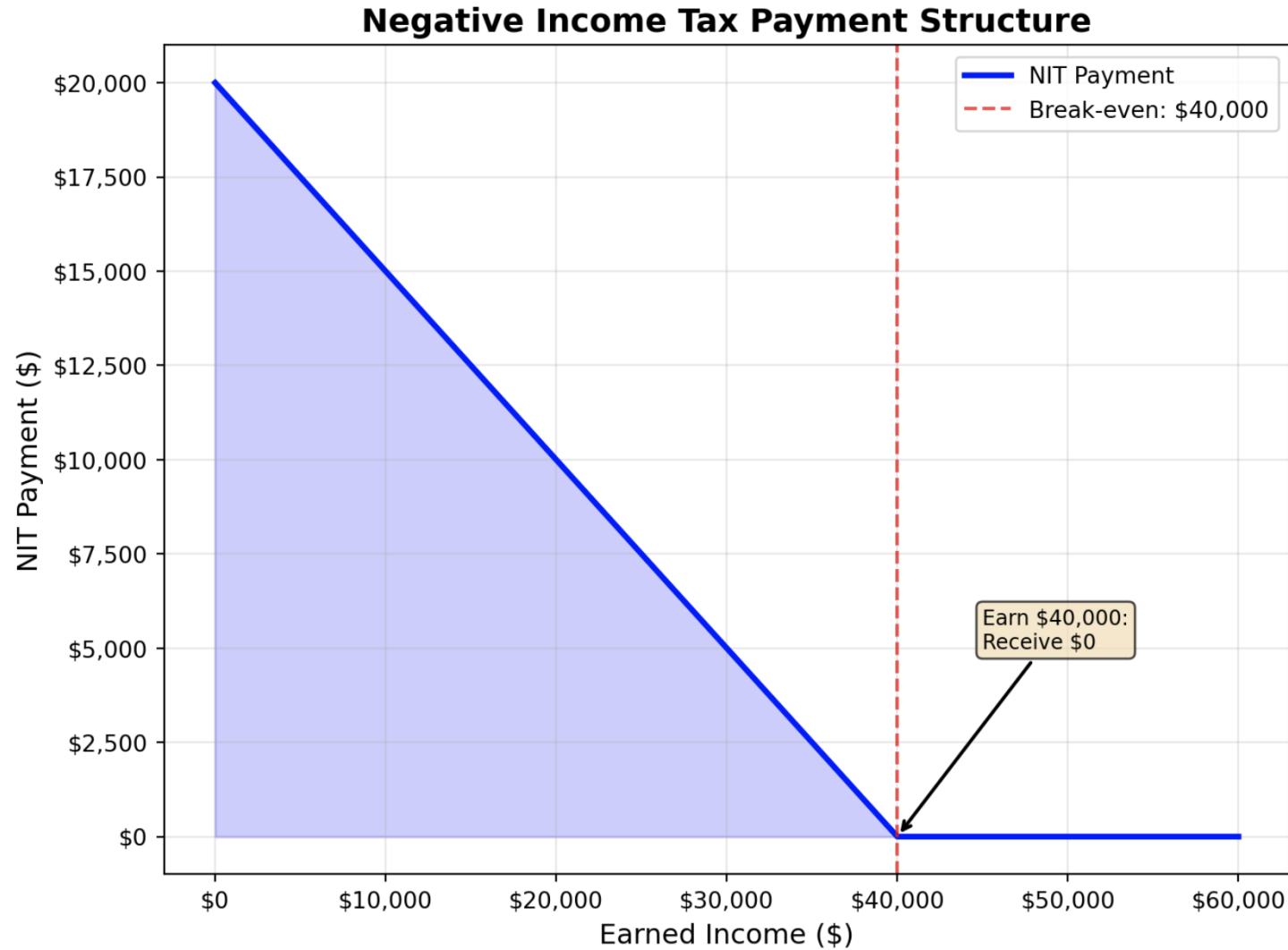
- Before Universal Basic Income (UBI), there was a policy proposal in the US in the 1960s known as Negative Income Tax (NIT)
- Economist Milton Friedman popularized the idea, claiming it would radically decrease complexity of welfare programs by replacing them with a lump sum payment
- In 1969, the Family Assistance Plan was introduced by the Nixon Administration: a negative income tax for working couples with children

Example of a Negative Income Tax

- Imagine the NIT guarantees 20,000, but for every dollar of earned income, the tax is 50 cents per dollar. Then your total benefit amount is equal to:

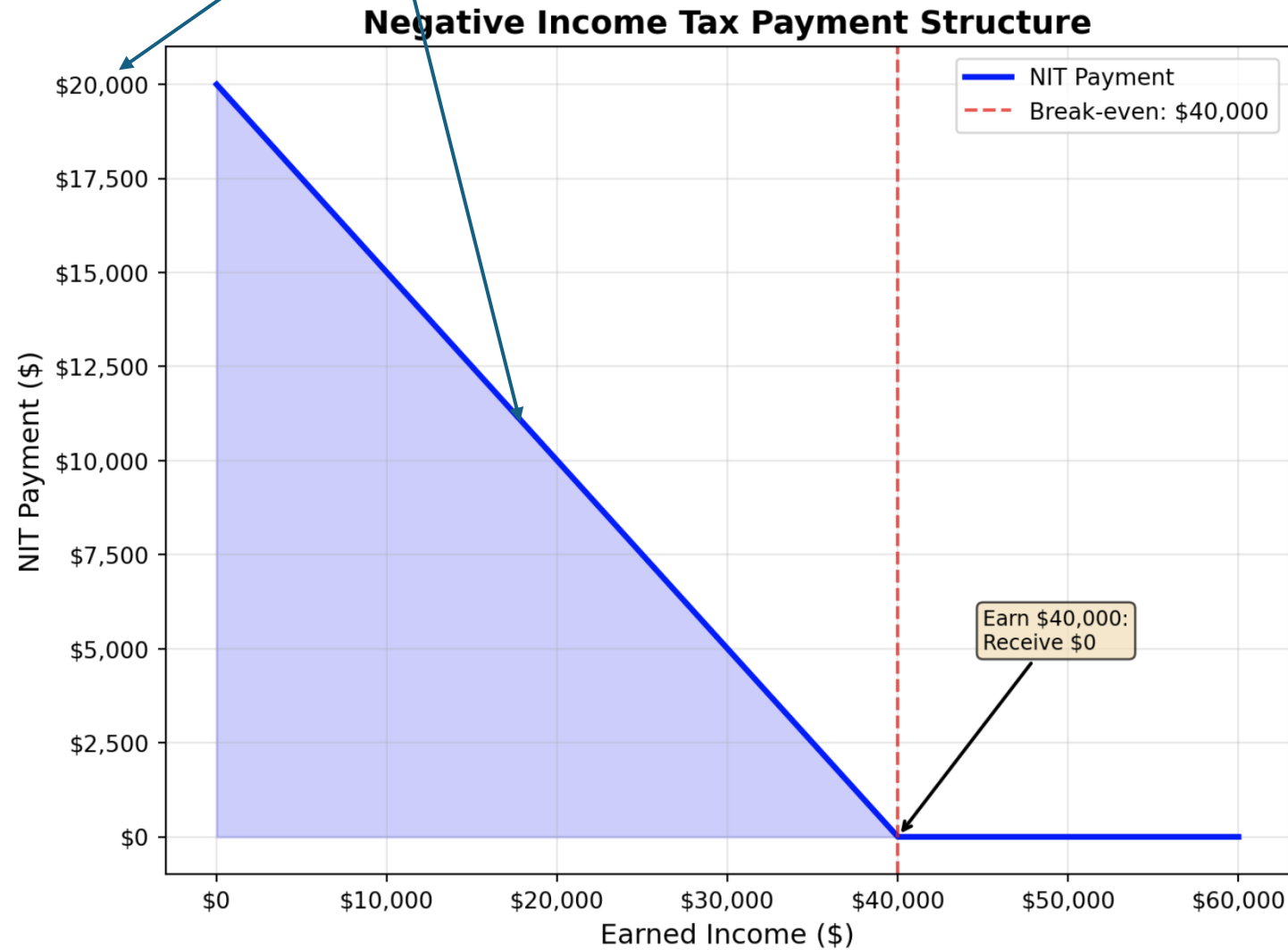
$$Benefit = 20,000 - 0.5 \cdot Earned\ Income$$

- Once you hit 40,000 dollars in income you hit the break-even point, at which point you receive zero benefit



Example of
a Negative
Income Tax

Experiments we study will vary a few parameters, such as guaranteed income amount or phase-out rate



Example of
a Negative
Income Tax

SIME/DIME

- The Seattle-Denver Income Maintenance Experiments were implemented in the 1970s
- Gave a large number of households a guaranteed income with various phase-out rates
- This is one of the first **field experiments** in Economics. A field experiment is an experiment (i.e. randomized treatment and control), but performed in a real-world setting rather than a lab setting

Table 1. Negative Income Tax Plans Tested in SIME/DIME

Initial Guarantee ⁽¹⁾	Tax Rate			
	50 percent	70 percent	70 percent (declining)	80 percent (declining)
\$3,800	X	X	X	X
\$4,800	X	X	X	X
\$5,600	X	X	—	X
Footnote: 1. 1971 dollars.				

SIME/DIME
Treatment
Groups

SIME/DIME Treatment Groups

In 2025 dollars, this is about \$30,000

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SIME/DIME
Treatment
Groups

SIME/DIME Treatment Groups

	Experimental Groups		Control Groups	
	Sample Size ⁽¹⁾	Average Transfer	Sample Size	Average Transfer
Year	Husbands			
1	1,393 (440)	\$1,361	1,071	\$263
2	1,294 (415)	\$1,296	990	\$220
3	1,183 (392)	\$1,276	851	\$210
4	(374)	\$1,233	804	\$195
5	(340)	\$990	562	\$165
Year	Wives			
1	1,408 (445)	\$1,455	1,089	\$295
2	1,343 (427)	\$1,524	1,026	\$327
3	1,253 (416)	\$1,564	895	\$383
4	(406)	\$1,618	849	\$397
5	(372)	\$1,323	600	\$348

SIME/DIME
Treatment
vs. Control
Gov't
Transfers

**B. Second year response, by NIT plan
(percentage difference in annual hours worked)**

	Tax Rate			
Guarantee	50 percent	70 percent	70 percent (declining)	80 percent (declining)
\$3,800	-6.7	-5.6	-10.0	-8.9
\$4,000 [\$4,800?]	-8.8	-1.5	-14.5	-9.9
\$5,600	-11.8	-10.4	—	-8.7

[SIME/DIME Labor Supply for Husbands](#)

**SIME/DIME
Impact on
Hours
Worked for
Husbands**

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SIME/DIME Labor Supply for Husbands

Size of transfer seems to matter for impact on labor supply

SIME/DIME
Impact on
Hours
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SIME/DIME Labor Supply for Husbands

Size of phase-out rate less consistent

SIME/DIME
Impact on
Hours
Worked for
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SIME/DIME Labor Supply for Husbands

	Tax Rate			
Guarantee	50 percent	70 percent	70 percent (declining)	80 percent (declining)
\$3,800	-24.7	-13.2	-1.6	-19.7
\$4,800	-29.3	-23.2	-20.7	-18.7
\$5,600	-28.1	-40.1	—	-12.6

SIME/DIME Labor Supply for Wives

SIME/DIME
Impact on
Hours
Worked for
Wives

SIME/DIME

- As theory predicts, there is a decrease in labor supply among individuals who receive the NIT that is increasing in the size of the payment
 - At the time, the effect was perhaps smaller than many economists predicted
 - However, perhaps large enough for opponents of negative income taxes to use it as a talking point
- The two sides of the argument never saw eye-to-eye. Some wanted more inclusive, higher-guaranteed income. Others wanted explicit work requirements to be eligible for the benefit

Universal Basic Income

- Universal Basic Income (UBI) is just like Negative Income Tax, but everyone receives the benefit
- There are a few talking points about these programs that can't necessarily be answered by our negative income tax experiments
 - NIT experiments only treated a subset of individuals in an economy. If everyone receives a payment, they may spend more, stimulating the economy and leading to employment growth
 - UBI is meant to be permanent. NIT experiments only had a short duration of a few years. Maybe individuals respond differently if they know the income is permanent

Universal Basic Income

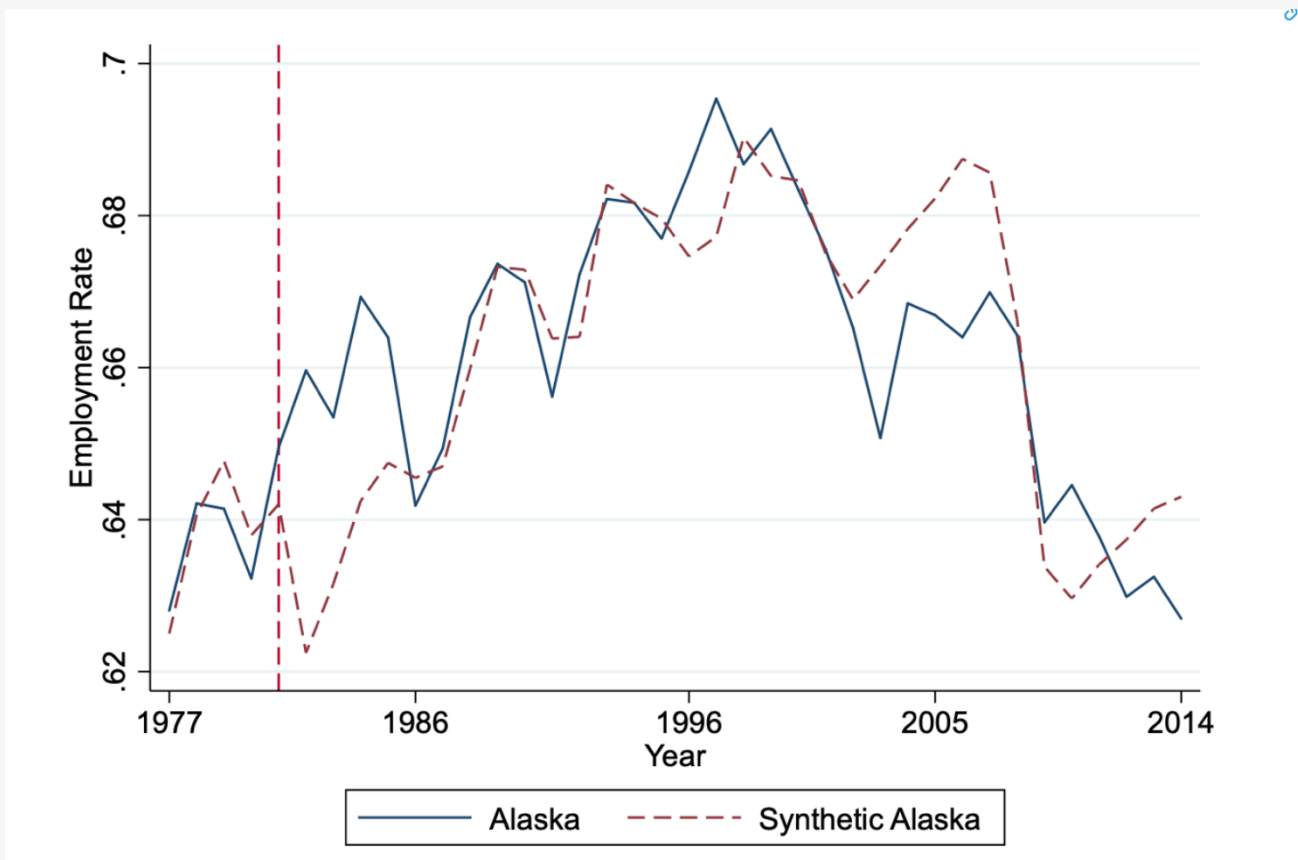
- There is limited (empirical) literature on Universal Basic Income, because we have few examples of it in the real world
- We are going to consider two studies
 - Alaska permanent fund: Alaska actually does have a UBI payment. It is far from providing basic needs, but is useful as a way to consider how non-labor income impacts labor supply decisions
 - GiveDirectly is part-way through a 12 year study that made UBI available for a random subset of villages in rural Kenya. This payment is large enough to cover basic needs

Alaska Permanent Fund

- Every year since 1982, Alaskan residents receive lumpsum payment from government
 - Even children are eligible for the payment
 - Average amount per household is around \$3,800. So doesn't realistically cover all basic needs, as proponents of UBI programs would generally argue for

Synthetic Control

- The authors use what is called a synthetic control to identify the impact of the Alaska Permanent Fund payment
 - They use data on employment rates across all states to find states that have similar employment trends to Alaska prior to 1982.
 - For example, if you average the employment rates of Utah and Wyoming, you get employment rates that look similar to Alaska
 - Therefore, you can use the average employment rate in Utah and Wyoming after 1982 to predict what would have happened in Alaska had they not passed the Alaska Permanent Fund



Synthetic Control

- No clear evidence that universal payment had any impact on employment rates
- Does not seem like Alaska Permanent Fund generates large drops in employment due to income effects

Banerjee et al. (2023)

- Banerjee et al. (2023) implement a Universal Basic Income Program in Rural Kenya by partnering with GiveDirectly (which was co-founded by Economics Professor at UCSD Paul Niehaus)
- Some impressive features of the paper's implementation of a UBI trial
 - Payment is enough to cover basic needs
 - They provide UBI for a randomly selected sample of villages. Everyone in the village receives the payment. In that sense it is universal
 - Payment is guaranteed for 12 years

Banerjee et al. (2023)

- Treatment arms. They tried a number of different versions of UBI
 - Long-term UBI: monthly payments for 12 years
 - Short-term UBI: monthly payments for 2 years
 - Lump Sum UBI: gives households an immediate lump-sum payment that is equivalent to the 2 years short-term payments
- Some of these choices were made specifically for the developing country context
 - Large lump sum payments can be important for starting business in this context

Banerjee et al. (2023)

Table 2: Overall Village Aggregates: Summed

	# Enterprises (1)	Revenues (2)	Costs (3)
Long Term Arm	9.93** [3.96]	61379.40** [24346.05]	32055.29* [16478.13]
Short Term Arm	3.39 [3.57]	23177.47 [16080.92]	8497.42 [10462.44]
Lumpsum Arm	14.67*** [3.92]	107746.75*** [34895.03]	71903.23*** [24360.84]

Banerjee et al. (2023)

Difference in Number of Enterprises in Treated Village that will receive 12 years of payments vs. control villages

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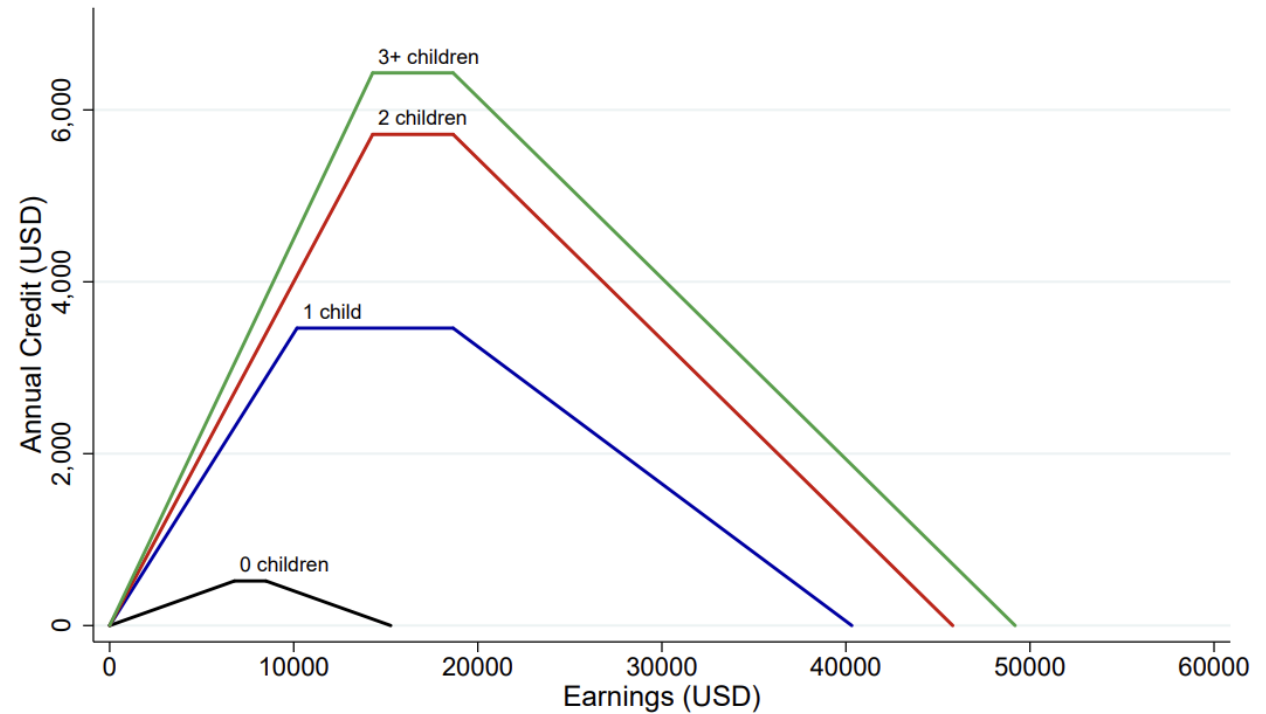
	Overall		Wage Employment		Self Employment (Non-Ag)	
	Hours (1)	Income (2)	Hours (3)	Income (4)	Hours (5)	Income (6)
Long Term Arm	61.92 [67.8]	503.40 [314.8]	-99.85** [46.45]	-275.53*** [105.3]	95.70* [54.74]	692.69** [284.59]
Short Term Arm	110.23 [77.59]	671.60* [365.13]	-68.73 [47.18]	323.03 [217.4]	90.72** [45.69]	215.67 [285.76]
Lumpsum Arm	79.42 [78.02]	1274.80** [571.56]	-27.30 [40.01]	272.42* [149.54]	81.79 [51.89]	875.18* [520.79]

The Earned Income Tax Credit (EITC)

- The EITC is the largest federal anti-poverty program in the United States. The way it works is that at low-income levels, wages are subsidized.
 - Provides about 60 billion USD to low-income families per year
 - To be eligible, you must have have earned income
 - Originally only available for individuals with children, but this was changed in the 1990s
 - The amount you can potentially receive in the credit depends on number of children and whether you are single vs. married

The Earned Income Tax Credit (EITC)

A: EITC Schedule in 2018



The Earned Income Tax Credit (EITC) Change Over Time (2 or more children)

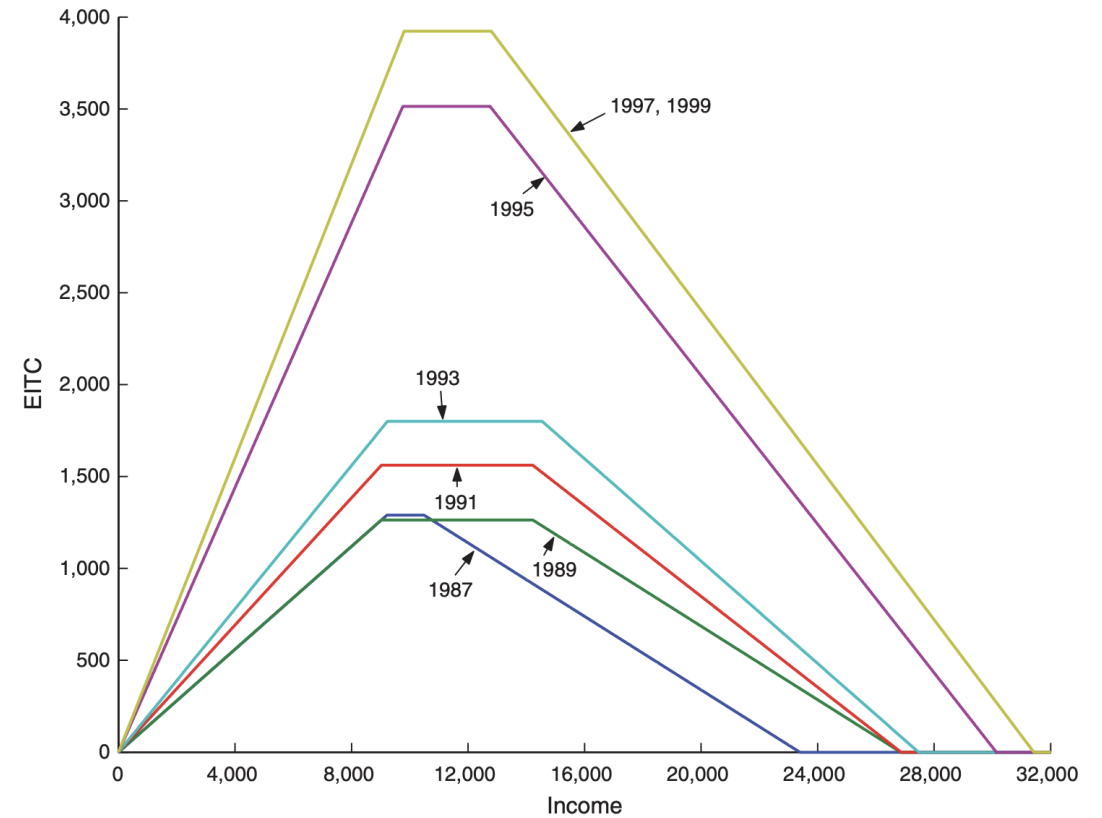
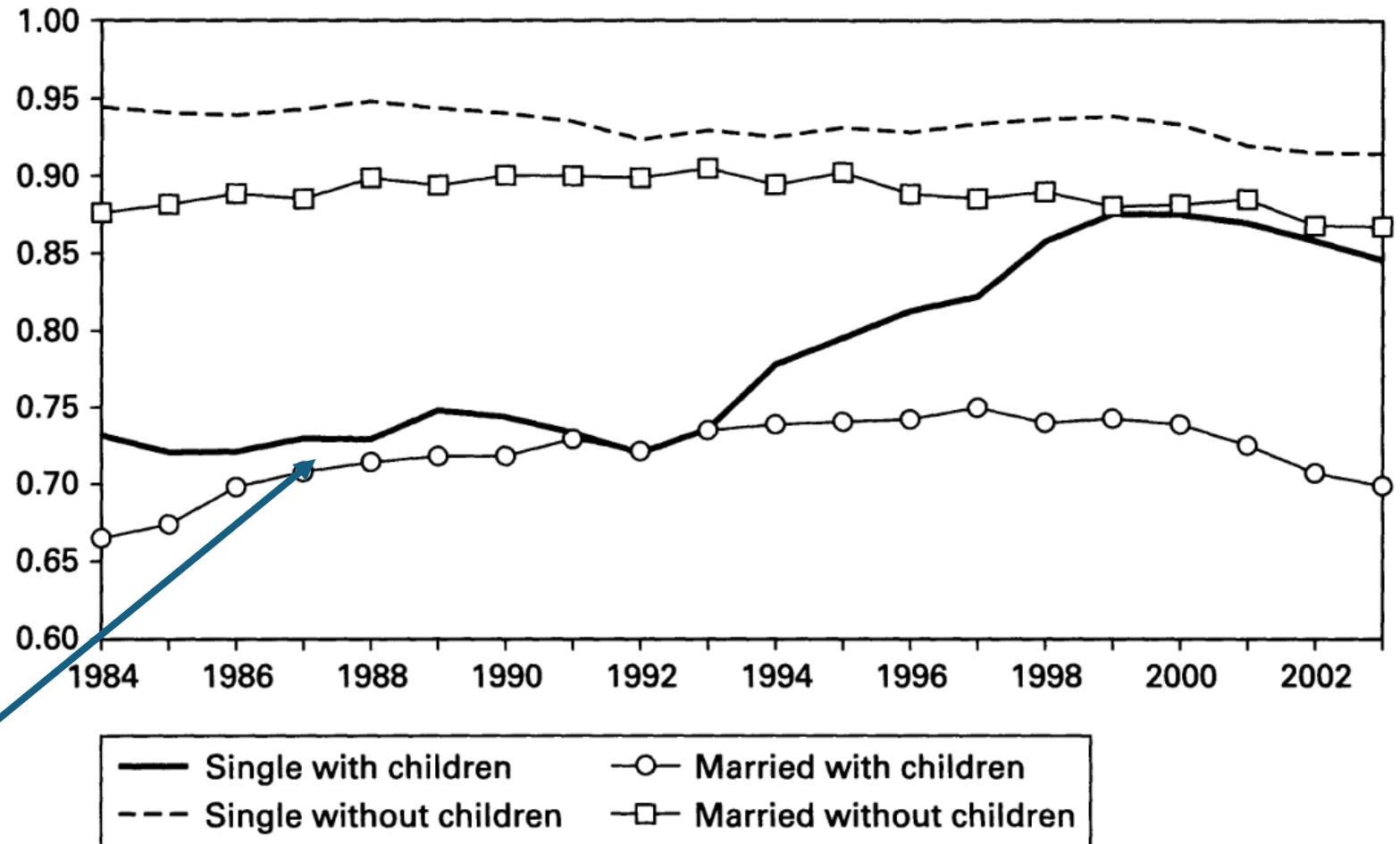


FIGURE 1. FEDERAL EITC SCHEDULES FOR FAMILIES WITH TWO OR MORE CHILDREN (Year 2000 dollars)

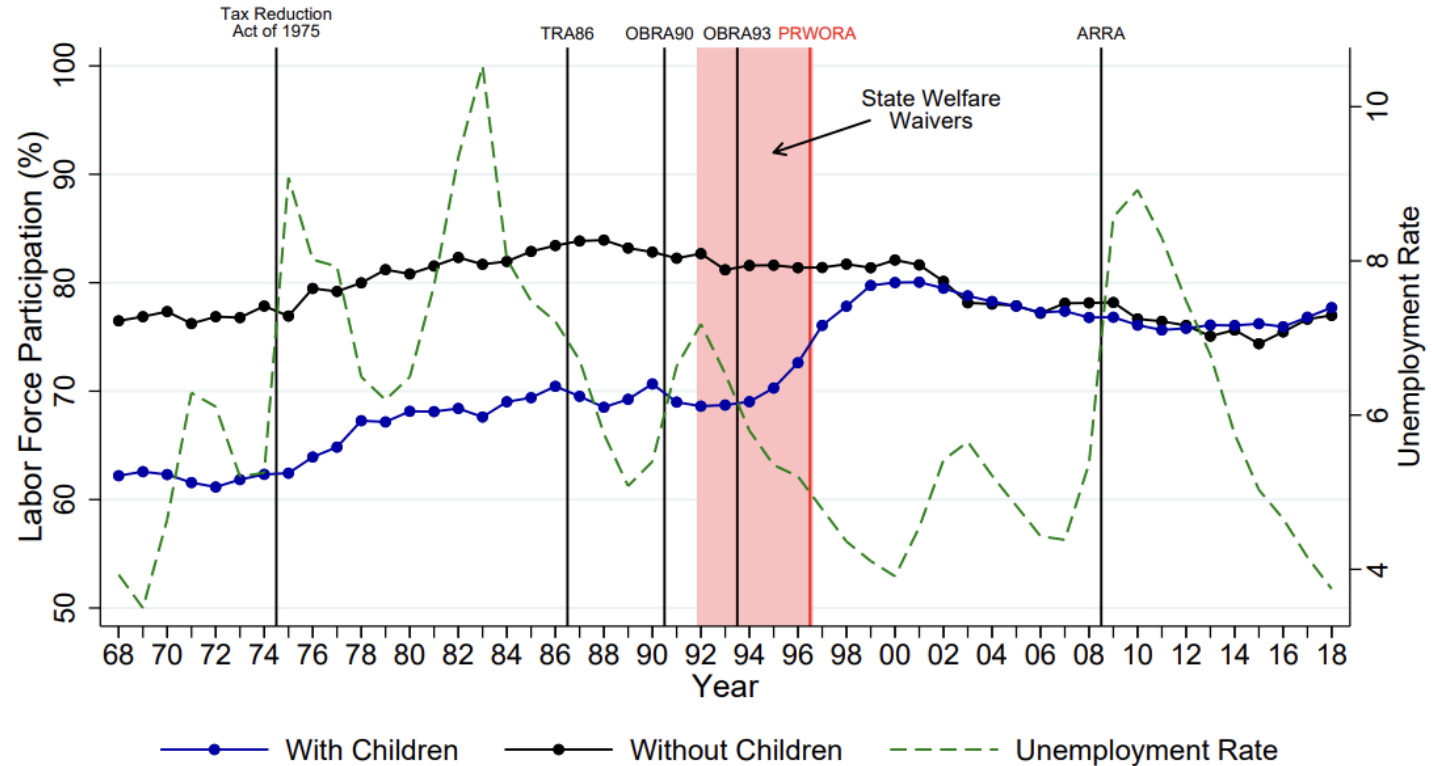
The Earned Income Tax Credit (EITC)



Many papers have studied employment effects of EITC expansions which started in late 1980s

The Earned Income Tax Credit (EITC)

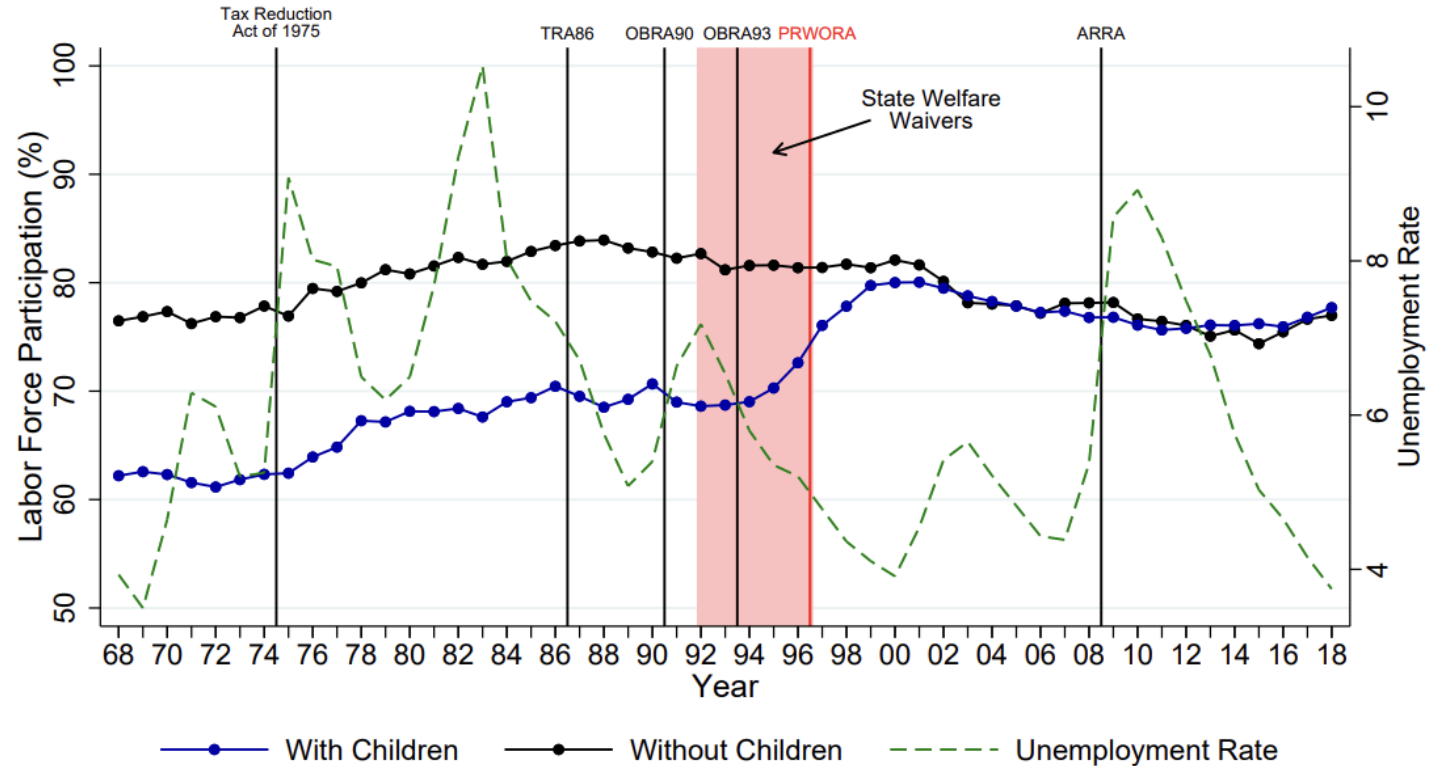
B: EITC Reforms and Confounders



Recent work (Kleven (2024)) has called into question the causal interpretation:
Finds that EITC expansions are correlated with other policies and macroeconomy

The Earned Income Tax Credit (EITC)

B: EITC Reforms and Confounders



Many states made changes to their welfare programs, which included new work requirements and time limits on receipt

Trends in Labor Force Participation

- This would be a good time to return to one of our big trends we discussed in week 1: the fall in labor force participation



Figure 9.15: Labor Force Participation Rate in the United States

Trends in Labor Force Participation

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Around 2007
LFP began to
fall

Figure 9.15: Labor Force Participation Rate in the United States

Trends in Labor Force Participation

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We will study
period between
2007 and 2015

Figure 9.15: Labor Force Participation Rate in the United States

Trends in Labor Force Participation

- We are going to discuss two broad reasons why the labor force participation rate may have started falling in this time period
 - Demographics: the labor force was aging around this time, so some of the decline is likely explained by demographic change
 - The value individuals place on leisure
 - While this may seem specific, you can think of anything that either makes work more difficult (i.e. pain, disability) or makes leisure activity more enjoyable (i.e. entertainment) as changing the value individuals place on leisure

Demographics

- Let's consider two groups of people
 - Individuals over the age of 55
 - Individuals between 25-54
- We are going to show how to form a prediction of the decline in labor force participation due to aging by considering the fraction of people that fall within these groups

Demographics

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Demographics

- In 2007, for individuals 25 and older
 - **35%** are above 55
 - **65%** are between 25-54
- Labor force participation rates are as follows:
 - Above 55 – **38%**
 - 25-54 – **83%**

$$LFP_{2007} = 0.35 \cdot 0.38 + 0.65 \cdot 0.83 = 0.673$$

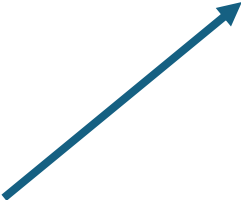
- Total labor force participation rate in 2007 is therefore:

Demographics

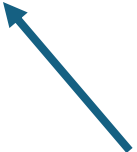
- Total labor force participation rate can be written as:

$$LFP_{2007} = 0.35 \cdot 0.38 + 0.65 \cdot 0.83 = 0.673$$

Individual above 55



Individuals 25-54



Question: what if labor force participation rates stayed the same, but the fraction above 55 grew

Demographics

Question: what if labor force participation rates stayed the same, but the fraction above 55 grew

$$LFP_{2015} = 0.41 \cdot 0.38 + 0.59 \cdot 0.83 = 0.646$$

2015 fraction above 55

2015 fraction 25-54

Demographics

Question: what if labor force participation rates stayed the same, but the fraction above 55 grew

$$LFP_{2015} = 0.41 \cdot 0.38 + 0.59 \cdot 0.83 = 0.646$$

2015 fraction above 55

2015 fraction 25-54

2007 participation rates

So ignoring any actual changes in participation rates, we would expect labor force participation to fall from 67.3% to 64.6% a **2.7 percentage point decline**

Demographics

Question: what if labor force participation rates stayed the same, but the fraction above 55 grew

Ignoring any actual changes in participation rates, we would expect labor force participation to fall from 67.3% to 64.6% a **2.7 percentage point decline**

In reality, from 2007 to 2015, labor force participation fell to 62.8%. So the decline from 64.6% to 62.8% is unexplained by aging

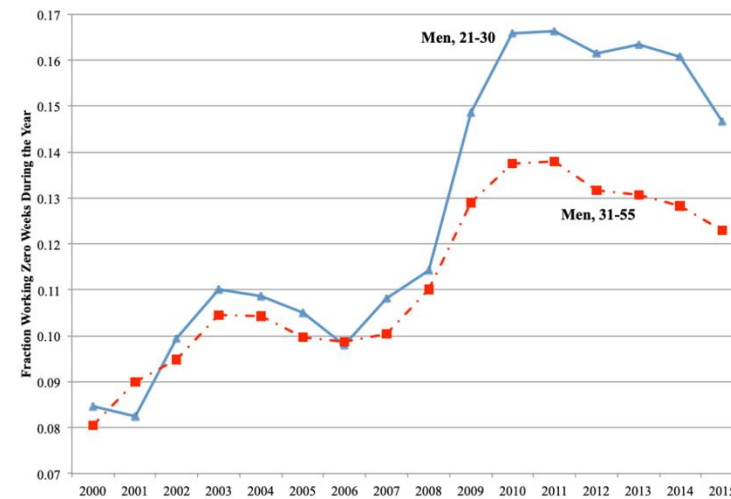
A More Thorough Decomposition

- Instead of assuming very broad age, we could have done this much more granularly by considering very narrow age bins (18-20, 21-23, etc.)
- Other work has done this, but found an overall similar finding: **About half of the fall in the labor force participation rate can be explained by aging**
- That leaves half to other reasons

The Great Recession

- One empirical finding that is at first surprising has been that the fall in labor force participation during the Great Recession (and after) is particularly large for young men

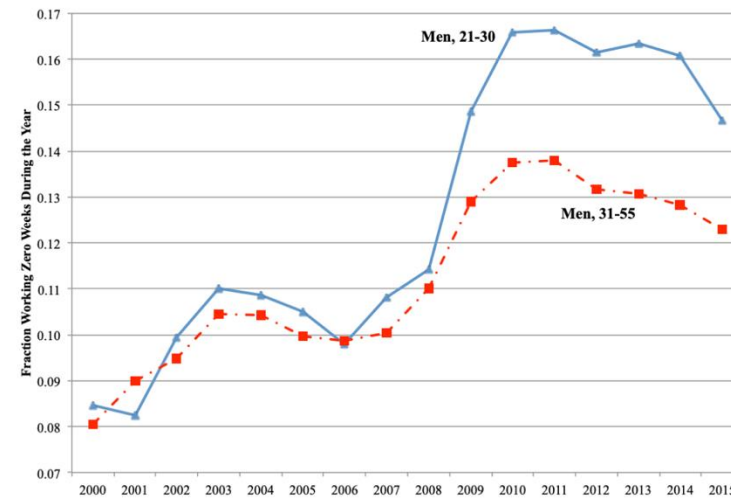
Figure 2: Fraction of Men With Zero Weeks Worked Over Prior Year by Age, March CPS



The Great Recession

- **Automation:** Cortes, Jaimovich, and Siu (2017) find that younger male workers were particularly impacted by automation of routine manual jobs
 - Therefore, automation can potentially explain a portion of this increase

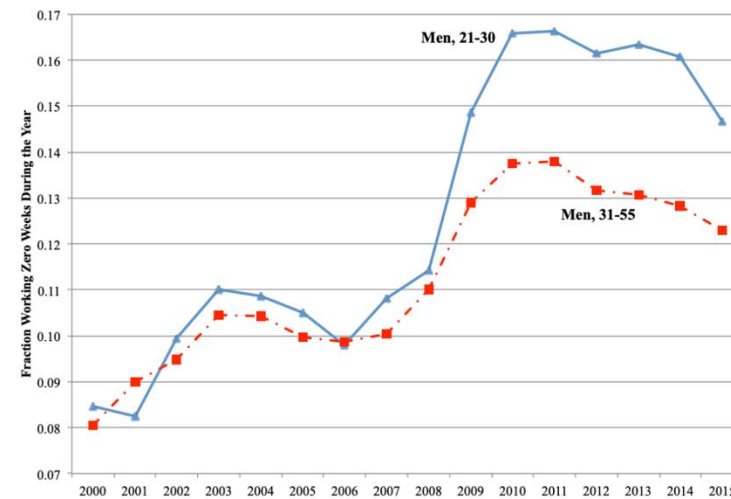
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The Great Recession

- The value of leisure: Aguiar et al. (2021) put forth a hypothesis that innovations to gaming technology have made leisure more valuable, leading to a drop in labor force participation
- Find evidence using time diaries of large increases in computer/game activity

Figure 2: Fraction of Men With Zero Weeks Worked Over Prior Year by Age, March CPS



Conclusion

- Main Concepts
 - Income effects vs. substitution effects
 - How individuals response to income and wage changes is a key input into policy design, but evidence is a bit mixed